

National University of Computer and Emerging Sciences
Lahore Campus

Signals and Systems

(EE2008)

Date: February 25, 2026

Course Instructor(s)

1. Dr. S. M. Sajid
2. Khalid Ijaz
3. Mohsin Yousuf (CM)

24L-6023

Roll No

48

Section



Student Signature

Sessional-I Exam

Total Time (Hrs): 1

Total Marks: 30

Total Questions: 3

Do not write below this line

1. Attempt all questions and remember to solve parts of the same question together.
2. Show all the steps with the help of labelled diagrams and mathematical equations.

CLO # 01: Apply fundamental concepts of signals and systems to classify and represent various types of signals and systems.

Q1: An input-output system is defined by a differential equation. The output for an input $x_1(t)$ is $y_1(t)$ and similarly for an input $x_2(t)$, the output is $y_2(t)$. [10]

$$x_1 \rightarrow y_1 \quad \text{and} \quad x_2 \rightarrow y_2$$

If the system is linear i.e. it obeys *superposition (homogeneity and additivity)*, show that:

$$\begin{aligned} &(D^N + a_1 D^{N-1} + \dots + a_{N-1} D + a_N) y(t) \\ &= (b_{N-M} D^M + b_{N-M+1} D^{M-1} + \dots + b_{N-1} D + b_N) x(t) \end{aligned}$$

Also, interpret values of M and N for practical systems.

CLO # 02: Analyze continuous and discrete-time Linear Time-Invariant (LTI) systems using convolution operations.

Q2: The unit impulse response of an LTIC system is [10]

$$h(t) = u(t - 3)$$

For this system, find and sketch its (zero-state) response $y(t)$ using *direct integration*, if the input is

$$x(t) = e^{-at} u(t), \quad a > 0$$

National University of Computer and Emerging Sciences
Lahore Campus

CLO # 02: Analyze continuous and discrete-time Linear Time-Invariant (LTI) systems using convolution operations.

Q3: Find and sketch the response $y(t) = x_1(t) * x_2(t)$ using graphical convolution for the following inputs. [10]

